

Iqbal Hidayat Rasyad

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Summary

Backend Engineer with strong focus on building and improving reliable systems using Golang. Experienced in optimizing backend services, implementing automation, and enhancing system reliability through clean architecture and observability-driven development.

Education

Information Technology

Universitas Gadjah Mada

Yogyakarta

Aug 2022 - Aug 2026

7th Semester Undergraduate Student, GPA 3.46/4

Experience

Software Developer Intern

PT BRI Asuransi Indonesia (BRINS)

Jan 2025 - Feb 2025

- Developed backend REST APIs using C# (.NET) with ASP.NET Core by applying OOP and SOLID principles
- Contributed to backend development using PHP (Lumen) for system monitoring applications with ELK Stack (Elasticsearch, Logstash, Kibana) and implemented replica for high availability and system observability
- Implemented frontend features using Angular and TypeScript and participated in UI/UX design using Figma
- Created Business Requirements Document (BRD) for new insurance product ideas to support business expansion initiatives

Projects

Comparative Analysis of Adaptive DevOps on Monolithic and Microservices Architectures

Undergraduate Thesis (In Progress)

- Designed and analyzed monolithic and microservices backends to study their impact on CI/CD pipeline design
- Developed an AI-augmented, rule-guided DevSecOps pipeline generator for adaptive CI/CD workflows
- Implemented Docker-based containerization and Kubernetes staging deployments for deployment validation
- Integrated observability readiness checks (health endpoints, service reachability, logging)

Event-Driven Order Processing System (Kafka & Golang)

- Designed an event-driven backend system using Apache Kafka
- Implemented producer and consumer services for asynchronous event processing
- Utilized topics, partitions, and consumer groups to improve scalability
- Deployed Kafka infrastructure using Docker Compose

DevOps Pipeline & Observability for Microservices

- Built Jenkins-based pipelines to automate build and deployment workflows for microservices
- Implemented centralized metrics collection and logging using Prometheus, Grafana, and Loki
- Monitored service health, resource usage, and logs in a distributed environment
- Focused on improving system reliability and observability-driven debugging

End-to-End Data Pipeline with Apache Airflow

- Built an end-to-end ETL pipeline integrating data from multiple external APIs
- Orchestrated scheduled data ingestion and transformation using Apache Airflow DAGs
- Persisted processed data into PostgreSQL for analytical use cases
- Containerized the pipeline to ensure reproducible and reliable execution

Smart Village Platform – Desa Pampang, Gunungkidul (KKN UGM 2025)

- Served as Public Relations Coordinator (Kormasit) in a 7-member team for 40 days, coordinating communication and documentation with village stakeholders
- Executed 7 individual programs and supported 30+ team programs within a 7-member KKN team during a 40-day community service program
- Developed a village tourism website with a lightweight CMS using Google Sheets
- Built a web-based IoT monitoring dashboard to track maggot growth data from mqtt
- Implemented backend APIs using Express.js and frontend interfaces with React
- Collaborated with IdeaHub developers to develop village tourism website as part of CSR project proposal for Telkomsel's BaktikuNegeriku program
- Delivered a maintainable system for non-technical users, community stakeholders, and IdeaHub

WhatsApp Reminder Chatbot (WAHA)

- Developed a chatbot service in Go to send scheduled reminders via messaging platforms
- Implemented message handlers, scheduling logic, and external API integration
- Focused on backend automation and reliable message delivery

Chat Backend Service

- Built a backend service for chat communication using Golang
- Implemented Redis as an in-memory data store for caching active conversations and user sessions
- Used Redis Pub/Sub to support real-time message delivery between chat participants
- Designed structured HTTP APIs for message handling and persistence
- Improved response time and system responsiveness through Redis-based caching

Cultural Guide Mobile App – Prambanan Temple

- Developed an Android application featuring interactive maps and location-based navigation
- Implemented geofencing logic to trigger content based on user proximity
- Integrated Firebase for authentication, data storage, and real-time synchronization
- Supported offline data access for improved user experience

IoT Monitoring System with MQTT & Web Dashboard (Capstone Project)

- Designed MQTT-based communication for real-time data exchange between IoT devices and backend services
- Implemented data publishing and subscription flows using MQTT for sensor and system status monitoring
- Developed a web-based dashboard using React to visualize real-time and historical IoT data
- Built backend services using Express.js to process MQTT messages and expose REST APIs
- Integrated MQTT data flow with web applications to support monitoring and analysis use cases

Academic Experience

Tutorial Assistant – Calculus Single Variable and Multi Variable

DTETI FT UGM

- Assisted students in understanding single and multivariable calculus concepts through guided discussions and tutorial sessions for ~30 students per class, conducted twice per week

Basic Science Practicum Assistant

DTETI FT UGM

- Assisted students in basic science experiments for ~50 students per session by utilizing Python-based software and relevant hardware tools

Skills

Technologies: Golang, TypeScript, JavaScript, Kubernetes, Kafka, Jenkins, Azure, C#, Docker

Languages: Indonesian (native), English (proficient)

Certificates

- Problem Solving - Basic (HackerRank)
- Perekayasaan Perangkat IoT (LSPTDI)
- ELT and ETL in Python (DataCamp)
- Dimensionality Reduction in Python (DataCamp)
- Advanced Deep Learning with Keras (DataCamp)
- Cleaning Data in Python (DataCamp)

Awards and Honors

The Most Integrious Member of OSIS

2020

MAN 4 Award